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Chain-of-Thought Hub: A Continuous Effort to Measure Large Language Models' Reasoning Performance

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https://github.com/FranxYao/chain-of-thought-hub

Abstract

As large language models (LLMs) are continuously being developed, their evaluation becomes increasingly important yet challenging. This work proposes Chain-of-Thought Hub, an open-source evaluation suite on the multi-step reasoning capabilities of large language models. We are interested in this setting for two reasons: (1) from the behavior of GPT and PaLM model family, we observe that complex reasoning is likely to be a key differentiator between weaker and stronger LLMs; (2) we envisage large language models to become the next-generation computational platform and foster an ecosystem of LLM-based new applications, this naturally requires the foundation models to perform complex tasks that often involve the composition of linguistic and logical operations. Our approach is to compile a suite of challenging reasoning benchmarks to track the progress of LLMs. Our current results show that: (1) model scale clearly correlates with reasoning capabilities; (2) As of May 2023, Claude-v1.3 and PaLM-2 are the only two models that are comparable with GPT-4, while open-sourced models still lag behind; (3) LLaMA-65B performs closely to code-davinci-002, indicating that with successful further development such as reinforcement learning from human feedback (RLHF), it has great potential to be close to GPT-3.5-Turbo. Our results also suggest that for the open-source efforts to catch up, the community may focus more on building better base models and exploring RLHF.

1. Introduction

Recently, the field of AI has been significantly impressed by the advances in large language models. LLMs exhibit multi-dimensional capabilities, and their evaluation is challenging. Generally, tuning a base language model into a chatbot is relatively easy, as demonstrated by the large variety of LLaMA-based (Touvron et al., 2023) models like Alpaca (Taori et al., 2023), Vicuna (Chiang et al., 2023), Koala (Geng et al., 2023), Dolly (Databricks, 2023), and so on. In a chitchat, all these models may perform superficially similarly to GPT-3.5-Turbo (Gudibande et al., 2023). At the current stage, the community is eager to know what are the key factors that clearly differentiate the better-performing models from the underperforming ones.

In this work, we consider the evaluation of complex reasoning. As noted by OpenAI (2023b), "In a casual conversation, the distinction between GPT-3.5 and GPT-4 can be subtle. The difference comes out when *the complexity of the task reaches a sufficient threshold*". A similar observation is made by the Google PaLM model family, as their developers discover that large models' chain-of-thought reasoning capability is clearly stronger than smaller models (Wei et al., 2022b;a). These observations indicate that the ability to perform complex tasks is a key metric.

The capability of performing complex reasoning is crucial for the models to become the next-generation computation platform. One example initiative is LangChain[1] where developers build applications powered by backend LLM engines, which generally require the model to perform complex tasks. Here the vision of pushing LLMs as the foundation of a new computational ecosystem also serves as a strong motivation to measure the models' reasoning performance.

To incentivize the research efforts in improving language models' reasoning performance, we propose the chain-of-thought hub (CoT Hub), a continuous open-source effort that tracks LLMs' reasoning capability using a carefully curated evaluation suite. CoT Hub is the first comprehensive comparison of very large LMs on reasoning benchmarks and currently consists of 19 major language models' (including the GPT, Claude, PaLM and LLaMA) performance on 6 benchmarks and more than 100 subtasks (including

[1]https://github.com/hwchase17/langchain

bi-lingual reasoning capabilities in Chinese), and we are continuously adding new models and datasets.

Observations made in CoT Hub shed light on many insights into LLM development: (1) the reasoning performance of LLMs highly correlates with models' scales; (2) as of May 2023, PaLM and Claude[2] are the only two model families that are comparable to (yet slightly worse than) the GPT model family; (2) LLaMA 65B (Touvron et al., 2023) the strongest open LLM to date, performs closely to code-davinci-002, the base model of GPT-3.5 family[3]. This indicates that **if aligned properly** (by doing supervised finetuning (SFT) and reinforcement learning from human feedback (RLHF) right) **LLaMA 65B can potentially improve further and perform on par with ChatGPT-3.5**. We hope our work gives meaningful guidance to the community's development of deployable LLMs.

2. Method

In this section we discuss the construction of Chain-of-Thought Hub. We first discuss our method for test data collection, then we discuss how we obtain the model performance on our test suite. Our main goal is to curate a high-quality collection of datasets that (1) is closely related to the actual usage of LLMs; (2) clearly differentiate the performance of stronger and weaker language models. We consider the following datasets:

GSM8k A widely used math reasoning datasets consisting of 8k problems that jointly test models' ability of arithmetic reasoning and composing math steps using language (Cobbe et al., 2021).

MATH A suite of challenging datasets consisting of 12k problems within 7 categories testing the models' advanced math and science reasoning. The problems in this dataset are very hard because they come from mathematics competitions written in Latex. Even GPT-4 has only 42.5% performance (Hendrycks et al., 2021).

MMLU An evaluation suite of 15k problems within 57 subjects testing model's high-school and college-level knowledge and reasoning (Hendrycks et al., 2020).

BigBench Hard A suite of language and symbolic reasoning tasks consisting 6.5k problems within 23 subsets that are particularly suitable for testing chain-of-thought prompting (Suzgun et al., 2022).

HumanEval A handwritten dataset of 164 Python programming problems with text comments and docstrings testing the models' coding ability (Chen et al., 2021).

C-Eval A Chinese evaluation suite for foundation models consisting of 13k multi-choice questions spanning 52 diverse disciplines and four difficulty levels (Huang et al., 2023).

We note that most of these datasets are already used in the evaluation of leading large language models, such as GPT-4 (OpenAI, 2023a) and PaLM-2 (Anil et al., 2023).

Few-Shot Chain-of-thought Prompting We use few-shot chain-of-thought prompting to evaluate LLMs. This marks a clear difference between our evaluation and the majority of other concurrent evaluations like HeLM (Liang et al., 2022), as most of them use answer-only prompting. We also emphasize that we use few-shot, rather than zero-shot prompting, because few-shot is a capability that exist in both pretrained and instruction-tuned checkpoints, v.s., zero-shot is more suitable for instruction-tuned checkpoints and may under-estimate the pretrained checkpoints.

Comparison to existing and concurrent work There are many great existing evaluation suites for large language models, such as HeLM, Chatbot Arena[4], and Open LLM Leaderboard[5]. The major difference between this work and these works are: (1) HeLM evaluates a significantly wider spectrum of tasks while we focus on evaluating reasoning. Most of the results from this work use chain-of-thought prompting (hence the name "Chain-of-Thought Hub") whereas HeLM mainly uses answer-only prompting (without CoT). (2) Chatbot Arena evaluate the dialog user preference we evaluate reasoning. (3) Open LLM Leaderboard focus on open-sourced LLMs, we jointly consider major LLMs, either open-sourced or not.

Using final answer accuracy as a proxy for reasoning capability Most of the datasets we consider share one pattern: to reach the final answer (either a number for math problems, a choice for multi-choice problems, or a fixed output for coding), the model needs to figure out the intermediate steps toward that answer. When evaluating, we only use the final answer accuracy but do not consider the correctness of intermediate steps. This is because empirically, the correctness of intermediate steps is strongly correlated with the final accuracy. If the intermediate steps are very wrong, the model is less likely to reach the final answer. If the final answer is correct, the intermediate steps are generally good enough (Wei et al., 2022b; Lewkowycz et al., 2022).

3. Experiments

First we discuss the model families we consider. We focus on the popular models in production, including GPT, Claude, PaLM, LLaMA, and T5 model families, specifically:

[2]https://www.anthropic.com/index/introducing-claude

[3]https://platform.openai.com/docs/model-index-for-researchers

[4]https://leaderboard.lmsys.org/

[5]https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard

Table 1. **Overall model performance on Chain-of-Thought Hub**. Numbers with an asterisk* are from our test scripts. For model types, base means the model checkpoint after pretraining, SIFT means supervised instruction finetuning. Others are from their corresponding papers. We observe: (1) there exist a gap between leading LLMs (GPT, Claude and PaLM) and open-source (LLaMA and FlanT5); (2) most leading LLMs are after RLHF, indicating the opportunity of improving open-sourced models using this technique; (3). model performance is generally correlated with model scale, indicating further opportunities in scaling, especially for open-source models. We further highlight that among open-sourced models, LLaMA 65B performs close to code-davinci-002, the base model of ChatGPT. This suggests that if RLHF is done right on LLaMA 65B, it may become close to ChatGPT.

Model	#Params	Type	GSM8k	MATH	MMLU	BBH	HumanEval	C-Eval
GPT-4	?	RLHF	92.0	42.5	86.4	-	67.0	68.7*
claude-v1.3	?	RLHF	81.8*	-	74.8*	67.3*	-	54.2*
PaLM-2	?	Base	80.7	34.3	78.3	78.1	-	-
gpt-3.5-turbo	?	RLHF	74.9*	-	67.3*	70.1*	48.1	54.4*
claude-instant-v1.0	?	RLHF	70.8*	-	-	66.9*	-	54.9*
text-davinci-003	?	RLHF	-	-	64.6	70.7	-	-
code-davinci-002	?	Base	66.6	19.1	64.5	73.7	47.0	-
Minerva	540B	SIFT	58.8	33.6	-	-	-	-
Flan-PaLM	540B	SIFT	-	-	70.9	66.3	-	-
Flan-U-PaLM	540B	SIFT	-	-	69.8	64.9	-	-
PaLM	540B	Base	56.9	8.8	62.9	62.0	26.2	-
text-davinci-002	?	SIFT	55.4	-	60.0	67.2	-	-
PaLM	64B	Base	52.4	4.4	49.0	42.3	-	-
LLaMA	65B	Base	50.9	10.6	63.4	-	23.7	38.8*
LLaMA	33B	Base	35.6	7.1	57.8	-	21.7	-
LLaMA	13B	Base	17.8	3.9	46.9	-	15.8	-
Flan-T5	11B	SIFT	16.1*	-	48.6	41.4	-	-
LLaMA	7B	Base	11.0	2.9	35.1	-	10.5	-
Flan-T5	3B	SIFT	13.5*	-	45.5	35.2	-	-

OpenAI GPT including GPT-4 (currently strongest), GPT-3.5-Turbo (faster but less capable), text-davinci-003, text-davinci-002, and code-davinci-002 (important previous versions before Turbo). See Fu & Khot (2022) for a comprehensive discussion.

Anthropic Claude including claude-v1.3 (slower but more capable) and claude-instant-v1.0 (faster but less capable)[6]. Strong competitor's GPT models.

Google PaLM including PaLM, PaLM-2, and their instruction-tuned versions (FLan-PaLM and Flan-U-PaLM). Strong base and instruction-tuned models.

Meta LLaMA including the 7B, 13B, 33B and 65B variants. Important open-sourced base models.

Google FlanT5 instruction-tuned T5 models demonstrating strong performance in the smaller model regime.

We report these models' performance on our CoT Hub suite. We note that due to the wide spectrum of the tasks and models we consider, the evaluation is nontrivial and even running inference takes effort. In addition, there are models that do not offer public access (such as PaLM), such that evaluating them is difficult. For these reasons, we report numbers using the following strategy: if the performance of a model is already reported in a paper, we refer to that paper; otherwise, we test them by ourselves. Note that this strategy is not comprehensive, as we still have a fraction of untested non-public models on some datasets. This is partly the reason we view our CoT Hub as a continuous effort.

Table 1 shows the overall results. We rank the models using GSM8k performance because it is a classical benchmark testing models' reasoning capabilities. Numbers marked by an asterisk are tested by ourselves, others are from the following sources: GPT-4 and PaLM-2 results are from their technical report (OpenAI, 2023a; Anil et al., 2023) respectively; GPT-3.5-Turbo's performance on HumanEval is also from OpenAI (2023a). Text-davinci-003, code-davinci-002 and text-davinci-002 performance are from the appendix in Chung et al. (2022) and from Fu et al. (2022). Minerva's performance is from Lewkowycz et al. (2022). PaLM's performance is from Chowdhery et al. (2022). Flan-PaLM and FlanT5 performance are from Chung et al. (2022). LLaMA's performance is from Touvron et al. (2023).

The gap between open-source and leading LLMs In

[6]https://console.anthropic.com/docs/api/reference

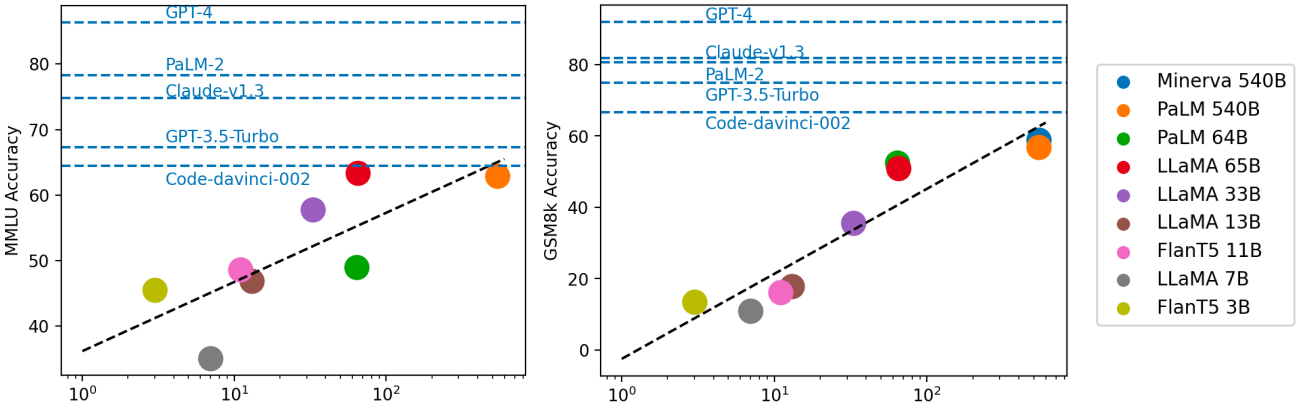

Figure 1. X-axis means the log of the model scale measured in billion parameters. We observe that model performance is generally correlated with scale, approximately showing a log-linear trend. Models without disclosing their scale generally perform better than models disclosing scale information. Our observations also indicate that the open-source community may still needs to explore/ figure out "the moat" about the scaling and RLHF for further improvements.

general, we observe a performance discrepancy between open-sourced models (like LLaMA and FlanT5) and close-sourced models (GPT, Claude and PaLM). Importantly, the performance of open-sourced models seems to be upper bounded by LLaMA 65B.

Leading LLMs are after RLHF We observe that except for PaLM-2, the top 6 models on the leaderboard are after reinforcement learning from human feedback. This strongly indicates the effectiveness of RLHF. Given that RLHF is still an underexplored area, we strongly encourage the community to study more on this topic.

Correlation between model scale and reasoning We further study the relationship between model scale and models' reasoning performance by visualizing model performance against model scale. Results are shown in Fig. 1. We see that: (1) generally, model performance is correlated with model scale, showing approximately a log-linear trend; (2) models that do not disclose their scale generally perform better than models that do, indicating that there is still a gap between open-source and close-source.

On the potential of LLaMA-65B Finally, we would like to highlight the impressive performance of LLaMA 65B. On MMLU it is close to code-davinci-002, the base model of GPT-3.5 series. On GSM8k, it is worse (presumably because it is not trained on code) but close and much better than other open-sourced models (presumably because it is trained to Chinchilla-optimal Hoffmann et al., 2022). Combining this observation with the fact that GPT-3.5-Turbo (ChatGPT) is an RLHF model based on code-davinci-002, **it may be possible to reproduce ChatGPT based on LLaMA 65B by applying the RLHF techniques** discussed in DeepMind Sparrow (Glaese et al., 2022) and Anthropic Claude (Askell et al., 2021; Bai et al., 2022a;b).

4. Conclusion and Future Work

In this work, we propose Chain-of-Thought Hub, an open-source, continuous effort to measure the reasoning capability of very large language models. Our results clearly show the performance differences between smaller and larger models, and between close-source and open-source models.

After carefully examining the results, we show two important directions for further improving open-sourced models: building better base models and exploring RLHF. We also point out the great potential of LLaMA 65B: if aligned properly by better SFT and RLHF, it could be possible to perform on par with ChatGPT-3.5.

In the future, we plan to further extend CoT Hub by: (1) including more carefully chozen reasoning datasets, especially datasets measuring commonsense reasoning, math theorem proving, and the ability to call outside APIs; (2) including more language models, such as LLaMA-based, instruction-finetuned models like Vicuna[7] and models through API access like Cohere[8] and PaLM-2 chat-bison-001[9]. (3) exploring methods for solving MATH, the probably most challenging datasets (recall that it consists of math- ematics competitions written in Latex), by calling APIs that compute symbolic and numerical calculus (like Wolfram Alpha[10]). In summary, we believe our work serves as an evaluation platform that guides the development of open-source large language models.

[7]https://lmsys.org/blog/2023-03-30-vicuna/
[8]https://cohere.com/generate
[9]https://cloud.google.com/vertex-ai
[10]https://www.wolframalpha.com/

References

Anil, R., Dai, A. M., Firat, O., Johnson, M., Lepikhin, D., Passos, A., Shakeri, S., Taropa, E., Bailey, P., Chen, Z., et al. Palm 2 technical report. *arXiv preprint arXiv:2305.10403*, 2023.

Askell, A., Bai, Y., Chen, A., Drain, D., Ganguli, D., Henighan, T., Jones, A., Joseph, N., Mann, B., DasSarma, N., et al. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*, 2021.

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022a.

Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., Chen, A., Goldie, A., Mirhoseini, A., McKinnon, C., et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022b.

Chen, M., Tworek, J., Jun, H., Yuan, Q., Pinto, H. P. d. O., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Chiang, W.-L., Li, Z., Lin, Z., Sheng, Y., Wu, Z., Zhang, H., Zheng, L., Zhuang, S., Zhuang, Y., Gonzalez, J. E., Stoica, I., and Xing, E. P. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, March 2023. URL https://lmsys.org/blog/2023-03-30-vicuna/.

Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.

Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, E., Wang, X., Dehghani, M., Brahma, S., et al. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2022.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Databricks. Free dolly: Introducing the world's first truly open instruction-tuned llm. Blog post, April 2023. URL https://www.databricks.com/blog/2023/04/12/dolly-first-open-commercially-viable-instruction-tuned-llm.

Fu, Yao; Peng, H. and Khot, T. How does gpt obtain its ability? tracing emergent abilities of language models to their sources. *Yao Fu's Notion*, Dec 2022. URL https://yaofu.notion.site/How-does-GPT-Obtain-its-Ability-Tracing-Emergent-

Fu, Y., Peng, H., Sabharwal, A., Clark, P., and Khot, T. Complexity-based prompting for multi-step reasoning. *arXiv preprint arXiv:2210.00720*, 2022.

Geng, X., Gudibande, A., Liu, H., Wallace, E., Abbeel, P., Levine, S., and Song, D. Koala: A dialogue model for academic research. Blog post, April 2023. URL https://bair.berkeley.edu/blog/2023/04/03/koala/.

Glaese, A., McAleese, N., Trebacz, M., Aslanides, J., Firoiu, V., Ewalds, T., Rauh, M., Weidinger, L., Chadwick, M., Thacker, P., et al. Improving alignment of dialogue agents via targeted human judgements. *arXiv preprint arXiv:2209.14375*, 2022.

Gudibande, A., Wallace, E., Snell, C., Geng, X., Liu, H., Abbeel, P., Levine, S., and Song, D. The false promise of imitating proprietary llms. *arXiv preprint arXiv:2305.15717*, 2023.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. Measuring mathematical problem solving with the math dataset. *NeurIPS*, 2021.

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. d. L., Hendricks, L. A., Welbl, J., Clark, A., et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

Huang, Y., Bai, Y., Zhu, Z., Zhang, J., Zhang, J., Su, T., Liu, J., Lv, C., Zhang, Y., Lei, J., et al. C-eval: A multi-level multi-discipline chinese evaluation suite for foundation models. *arXiv preprint arXiv:2305.08322*, 2023.

Lewkowycz, A., Andreassen, A., Dohan, D., Dyer, E., Michalewski, H., Ramasesh, V., Slone, A., Anil, C., Schlag, I., Gutman-Solo, T., et al. Solving quantitative reasoning problems with language models. *arXiv preprint arXiv:2206.14858*, 2022.

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023a.

OpenAI. Gpt-4, 2023b. URL https://openai.com/research/gpt-4.

Suzgun, M., Scales, N., Schärli, N., Gehrmann, S., Tay, Y., Chung, H. W., Chowdhery, A., Le, Q. V., Chi, E. H., Zhou, D., et al. Challenging big-bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*, 2022.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., Yogatama, D., Bosma, M., Zhou, D., Metzler, D., et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022a.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., and Zhou, D. Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*, 2022b.